Givi Tsvariani

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Education

Rutgers University New Brunswick, NJ

Mathematics & Computer Science Major, Physics Minor. 3.2/4.0. *Sep. 2024 – Expected Jun. 2028*

- **Computer Science**: Principles of Programming Languages, Systems Programming, Computer Architecture.
- **Mathematics**: Graph Theory, Abstract Algebra I, Linear Optimization, Differential Equations.
- **Physics**: Foundations of Quantum Mechanics, Analytical Physics, Quantum Mechanics and Atomic Physics

Experience

Intern Jun. 2024 – Aug. 2024

D&J Electronics *Burlington, NJ*

- Diagnosed and resolved hardware and operating system issues on company laptops and servers.
- Configured and re-imaged workstations for large-scale (100+ machines) deployment at conferences.

Open Source Maintainer Jan. 2022 – Present

Independent *Remote*

- Maintainer of multiple packages in the world's largest package repository, Nixpkgs.
- Developer and maintainer of the Nixp Lisp dialect as an alternative Nix frontend in the Odin language. This also brings a type-safe experience to Nix.
- Created and maintain Nix flakes such as: Nixtants (free-form constants) and mkScript (Nix script management).

Projects

Ysyldur | *Linux, Systemd, Nix, BASH* [GitHub]

- An importable Nix flake for setting up and managing the Linux kernel.
- Designed with a clean API for setting kernel variant, version, parameters, sysctl settings, drivers, and swapping kernel modules.
- Relies on Nixpkgs and Chaotic-Nyx repositories for kernel packages.
- Currently running in production on my fleet of over 10 NixOS hosts (laptops, servers, VMs, and VPSs).

Loid | *Rust, TypeScript, JavaScript* [GitHub]

- Working with a team on the Loid programming language (WiP) and its ecosystem.
- Uses a dependent type system, first class types, functions, and OOP features while allowing for nicer side effects.
- Written in Rust with the purpose of compiling to JavaScript code for both browser and node targets.
- Uses dependent type theory (Martin-Löf), the Call-By-Push-Value paradigm (inspired, not directly), and monadic error handling for compile-time type-safety.
- Designed with DSL creation and Web Development in mind, featuring a huge standard library with dozens of separate modules.

Acrithis | *Rust, SQLite, Nushell, Ollama, Nix, Lean4* [GitHub]

- A Directed Acyclic Graph (DAG) enforcing SQLite wrapper database for Lean4 programs.
- Sets up a Merkle Tree structure for theorem and axiom dependencies for each program.
- AI agents use Acrithis as an immutable append and read-only database for large mathematical proofs.
- When in use for Mathematics proofs, expected to eliminate up to 20% of hallucinations.
- Written in Rust, as a part of my final project paper for the AI in Mathematics class at Rutgers (Math496).

Technical Skills

Programming Languages: TypeScript, Python, JavaScript, Rust, Odin, Go, Gleam, Racket.
Developer Tools: Git, Docker, Webpack, Linux, GitHub/Forgejo, CI/CD, Postgres, Typst, LaTeX.
Libraries/Frameworks: jQuery, ExpressJS, AstroJS, Lustre, Ramda, fp-ts, MochaJS, ChaiJS, WPILib.
Systems: Ubuntu/Debian, NixOS, GNU Coreutils, BASH & Nushell, AWS EC2, Hetzner.
Computer Science: Functional Programming, Compilers, Type Systems, Formal Verification, Package Management.
Mathematics: Graph Theory, Type Theory (Martin-Löf), ZFC Set Theory, Lambda Calculus, Category Theory.
Languages: English, Georgian, conversational Spanish.